International Journal of Engineering and Information Technology
Vol 2 , No. 1
ISSN 0975-5292 (Print)
ISSN 0976-0253 (Online)
IJEIT 2010, 2(1), 60-63

CLOUD COMPUTING: A STUDY OF INFRASTRUCTURE AS A SERVICE (IAAS)

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Abstract: Cloud computing is an increasingly popular paradigm for accessing computing resources. In practice, cloud service providers tend to offer services that can be grouped into three categories: software as a service, platform as a service, and infrastructure as a service. This paper discuss the characteristics and benefits of cloud computing. It proceeds to discuss the Infrastructure as a service (IaaS). This paper aims to provide a means of understanding and investigating IaaS. This paper also outlines the responsibilities of IaaS provider and the facilities to IaaS consumer

Keywords: Cloud Computing, IaaS, PaaS, SaaS.

Introduction

Cloud computing is a way of referring to the use of shared computing resources, and it is an alternative to having local servers handle applications. Cloud computing groups together large numbers of compute servers and other resources and typically offers their combined capacity on an on-demand, pay-per-cycle basis. The end users of a cloud computing network usually have no idea where the servers are physically located—they just spin up their application and start working. According to Wang and von Laszewski [1], Cloud computing can be defined as "*A computing Cloud is a set of network enabled services, providing scalable, QoS guaranteed, normally personalized, inexpensive computing platforms on demand, which could be accessed in a simple and pervasive way*". According to GuiyiWei et al [2] *Cloud computing is a natural evolution for data and computation centers with automated systems management, workload balancing, and virtualization technologies.* Cloud-based services integrate globally distributed resources into seamless computing platforms. Recently, a great deal of applications are increasingly focusing on third-party resources hosted across the Internet and each has varying capacity. Fig. 1 show the logical diagram of cloud computing.

It is a paradigm shift (change in a fundamental model of events) following the shift from mainframe to client-server. It is a paradigm shift (change in a fundamental model of events) following the shift from mainframe to client-server. Details are abstracted from the users who no longer have need of, expertise in, or control over the technology infrastructure "in the cloud" that supports them[3]. Cloud computing describes a new supplement, Details are abstracted from the users who no longer have need of, expertise in, or control over the technology infrastructure "in the cloud" that supports them [3]. consumption and delivery model for IT services based on the Internet, and it typically involves the provision of dynamically scalable and often virtualized resources as a service over the Internet [4,5] It is a byproduct and consequence of the ease-of-access to remote computing sites provided by the Internet [5].

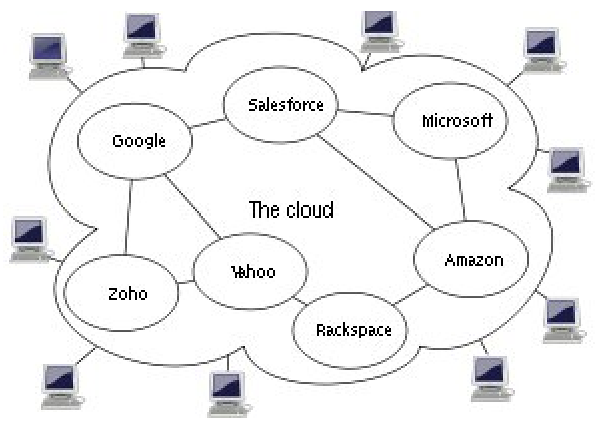

Fig. 1. Logical diagram of cloud computing

The term *cloud* is used as a metaphor for the Internet, based on the cloud drawing used in the past to represent the telephone network [6], and later to depict the Internet in computer network diagrams as an abstraction of the underlying infrastructure it represents [7]. Typical cloud computing providers deliver common business applications online which are accessed from another web service or software like a web browser, while the software and data are stored on servers.

A technical definition is a computing capability that provides an abstraction between the computing resource and its underlying technical architecture (e.g., servers, storage, networks), enabling convenient, on-demand network access to a shared pool of configurable computing resources that can be rapidly provisioned and released with minimal management effort or service provider interaction." This definition states that clouds have five essential characteristics [8]:

- on-demand self-service,
- broad network access,
- resource pooling,
- rapid elasticity, and
- measured service.

Benefits of Cloud Computing

- Cloud technology is paid incrementally, saving organizations money.
- Organizations can store more data than on private computer systems.
- No longer do IT personnel need to worry about keeping software up to date.
- Cloud computing offers much more flexibility than past computing methods.
- Employees can access information wherever they are, rather than having to remain at their desks.
- No longer having to worry about constant server updates and other computing issues, government organizations will be free to concentrate on innovation.
- Decoupling and separation of the business service from the infrastructure needed to run it (virtualization).
- Flexibility to choose multiple vendors that provide reliable and scalable business services, development environments, and infrastructure that can be leveraged out of the box and billed on a metered basis—with no long term contracts
- Elastic nature of the infrastructure to rapidly allocate and de-allocate massively scalable resources to business services on a demand basis.
- Cost allocation flexibility for customers wanting to move CapEx into OpEx
- Reduced costs due to operational efficiencies, and more rapid deployment of new business services

Principles of cloud computing

- Virtualization and automation
- Interchangeable (fungible) resources such as servers, storage and network
  - Management of these resources as a single fabric
  - Elastic capacity (scale up or down) to respond to business demands
- Applications (and the tools to develop them) that can truly scale out

2 Architectural layers of Cloud Computing

Cloud computing is typically divided into three levels of service offerings as showed in Fig. 2: Software as a Service (SaaS), Platform as a Service (PaaS), and Infrastructure as a service (IaaS). These levels support virtualization and management of differing levels of the solution stack.

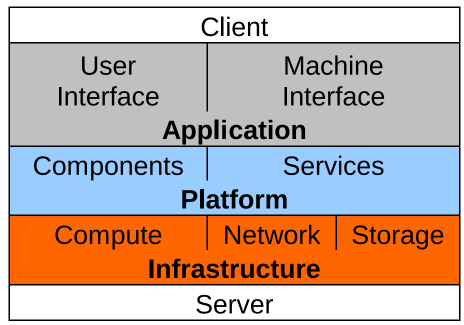

Fig. 2 Cloud Computing Stack [9]

2.1 Software as a Service

This is the idea that someone can offer you a hosted set of software (running on a platform and infrastructure) that you don't own but pay for some element of utilization - by the user, or some other kind of consumption basis. Here you don't have to do any development or programming, but you may need to come in and configure the (very flexible, configurable and sometimes customizable) software. You don't have to purchase anything. You just pay for what you use. A SaaS provider typically hosts and manages a given application in their own data center and makes it available to multiple tenants and users over the Web. Some SaaS providers run on another cloud provider's PaaS or IaaS service offerings. Oracle CRM on Demand, Salesforce.com, and Netsuite are some of the well known SaaS examples.

2.2 Platform as a Service

This is the idea that someone can provide the hardware (as in IaaS) plus a certain amount of application software - such as integration into a common set of programming functions or databases as a foundation upon which you can build your application. Platform as a Service (PaaS) is an application development and deployment platform delivered as a service to developers over the Web. It facilitates development and deployment of applications without the cost and complexity of buying and managing the underlying infrastructure, providing all of the facilities required to support the complete life cycle of building and delivering web applications and services entirely available from the Internet. This platform consists of infrastructure software, and typically includes a database, middleware and development tools. A virtualized and clustered grid computing architecture is often the basis for this infrastructure software. Some PaaS offerings have a specific programming language or API. For example, Google AppEngine is a PaaS offering where developers write in Python or Java. EngineYard is Ruby on Rails. Sometimes PaaS providers have proprietary languages like force.com from Salesforce.com and Coghead, now owned by SAP.

2.3 Infrastructure as a Service

Infrastructure as a Service (IaaS) is the delivery of hardware (server, storage and network), and associated software (operating systems virtualization technology, file system), as a service. It is an evolution of traditional hosting that does not require any long term commitment and allows users to provision resources on demand. Unlike PaaS services, the IaaS provider does very little management other than keep the data center operational and users must deploy and manage the software services themselves just the way they would in their own data center. Amazon Web Services Elastic Compute Cloud (EC2) and Secure Storage Service (S3) are examples of IaaS offerings.

3 Understanding Infrastructures as a Service (IaaS)

Infrastructure as a Service is a form of hosting. It includes network access, routing services and storage. The IaaS provider will generally provide the hardware and administrative services needed to store applications and a platform for running applications. Scaling of bandwidth, memory and storage are generally included, and vendors compete on the performance and pricing offered on their dynamic services. The service provider owns the equipment and is responsible for housing, running and maintaining it. IaaS can be purchased with either a contract or on a pay-as-you-go basis. However, most buyers consider the key benefit of IaaS to be the flexibility of the pricing, since you should only need to pay for the resources that your application delivery requires.
Characteristics and components of IaaS include:

- Utility computing service and billing model.
- Automation of administrative tasks.
- Dynamic scaling.
- Desktop virtualization.
- Policy-based services.
- Internet connectivity.

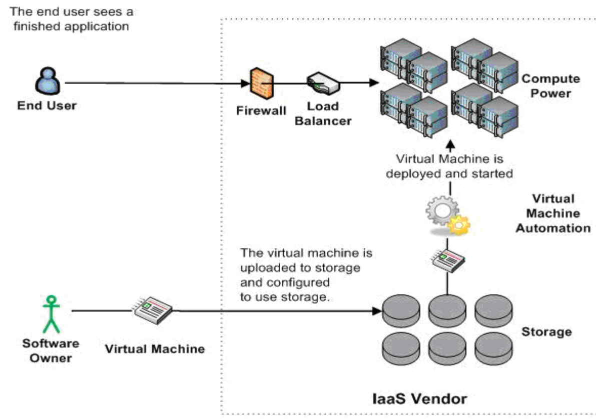

Figure 3 – Infrastructure as a Service [10]

IaaS provides an environment for running user built virtualized systems in the cloud. Figure 3 illustrates how a virtual machine is built for an IaaS environment, uploaded to the environment, configured, and then deployed within the environment. Using this technique virtual machines are created on premise and loaded with all the software that will eventually run in the cloud. This includes custom built software as well as licensed software. After the virtual machine is built it is uploaded to the IaaS vendor’s hosting environment where it can be configured to use the IaaS vendor’s raw storage. Once configured, the virtual machine can be deployed and started via some form of automation which automatically finds available hardware to run the virtual machine. Once the virtual machine is started the IaaS vendor can ensure that the running virtual machine continues to look healthy as a whole. The computers needed to run the application and the raw storage that is needed by the application are owned and supported by the IaaS vendor. It is the responsibility of the customer to monitor all the custom built software and licensed software to insure that they are operating properly. IaaS is an option that is very flexible and is the best choice for moving applications to the cloud when there is no time to rework the application’s code for a cloud environment.

3.1 IaaS provider and Consumer

The key roles in a cloud environment include the service consumer and the service provider. The cloud service consumer needs a secure anytime anywhere access to low cost services that are flexible and easy to use. The biggest hurdle to adoption of cloud has to do with consumers discomfort in the following areas: security of service and the underlying data, service availability and reliability, service management to ensure service level agreements, ensuring control over access and policies, and the appropriate administration to facilitate flexible pricing structures [11]. The service provider actually runs the service that the service consumer wants and was designed and developed by the service creator. The cloud services creator needs tools and capabilities to offer differentiated services, offer incentives to ensure that consumers keep coming back to use the services, and the ability to change services on-demand to stay competitive and address threats. The service provider needs their IT resources integrated so their usage is optimized, the ability to add/remove resources on demand, a non-disruptive way to save money, and the means to charge for usage.

3.2 Consumer's view on IaaS

- Enable users to access applications from anywhere
- A modular system, which is flexible, scalable, virtualized and automated.
- Resilient and always available
- Enable to put applications and data on platform provisioning & maintenance by provider
- Own the hardware & nuances about provisioning & maintaining the OS & hygiene facts like space and power etc

3.3 Provider's view on IaaS

- Provide virtual infrastructure (server, storage and Network virtualization).
- Responsible for provisioning of space, power & cooling.
- Deploy web based applications to easily provision infrastructure for customer on demand.
- Responsible to provide load balancing services.
- Eases the process of cloning apps on additional infrastructure instances.
- Service level agreements with customers on "availability of infrastructure services"
- In a dense, shared, and pooled environment, the security of CPUs, data, and network is paramount.
- Account Management & Provisioning.

4 Conclusion

To summarize, an Infrastructure as a Service (IaaS) offering provides solid cost savings because the infrastructure associated with providing compute power, storage, and networking does not need to be purchased and maintained by the customer. These assets are the responsibility of the IaaS vendor and customers are only charged for what they use when they use it. Table 1 summarizes Infrastructure as a Service [9]. IaaS is also a flexible offering that often appeals to infrastructure architects. Infrastructure architects like IaaS because it provides an infrastructure based approach to outsourcing datacenter workloads to the Cloud. If an application can be virtualized it can be uploaded to an IaaS environment and run.

Table 1: **IaaS Summary**

Offering	Compute power, storage, and networking infrastructure. Some IaaS vendors may also provide Cloud Services.
Unit of deployment	Virtual Machine Image
Pricing structure	Compute usage per hour, data transfer in/out per GB, IO requests per million, storage per GB, data transfer in/out to storage per GB, data storage requests per thousand. All charges per billing period.
Customer	Software owner that would like an application hosted in the internet for their end users.
Examples	Amazon, GoGrid, and Rackspace.

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